

Fingerprint Enhancement Report

Report-ready evidence for visual inspection, quantitative comparison and reproducible algorithm evaluation.

Algorithm	RHLT
Input file	1.png
Pipeline build	rhlt-primary-quality-adaptive-v3.2
Generated	2026-09-07 11:03:22 Malay Peninsula Standard Time

Enhancement Quality Summary

Metric	Value	Interpretation
Contrast Improvement (CII)	1.26x	≥ 1.0 = improved
Sharpness Improvement	+56.8%	Positive = sharper ridges
Ridge-Valley Clarity Change	+58.2%	Positive = cleaner ridge/valley separation
SSIM (Structure Preserved)	0.980	≥ 0.80 = well preserved
Mean Orientation Coherence	0.573	Closer to 1.0 = more parallel ridge flow
Minutiae Detected	298	Ridge endings + bifurcations on skeleton

RHLT Candidate Decision

Final selected output: **Proposed Improved RHLT**

Proposed Improved RHLT was selected as the primary method: it passed all structural safety checks and its score difference was +0.0027 (allowed regression tolerance 0.0100).

Recorded RHLT Parameters

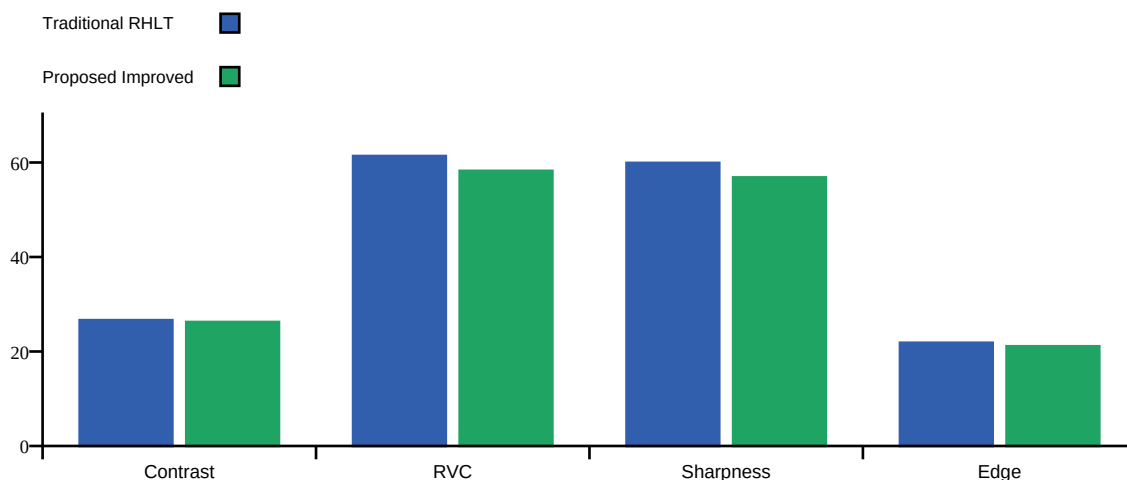
Parameter	Recorded value
RHLT PSF size	65
Topological charge	1
Orientation block size	16
Orientation bins	12
Gabor kernel size	17
Gabor wavelength	10.000
Maximum Gabor weight	0.450
Frequency block size	24
Selection tolerance	0.010

Traditional vs Proposed Quantitative Comparison

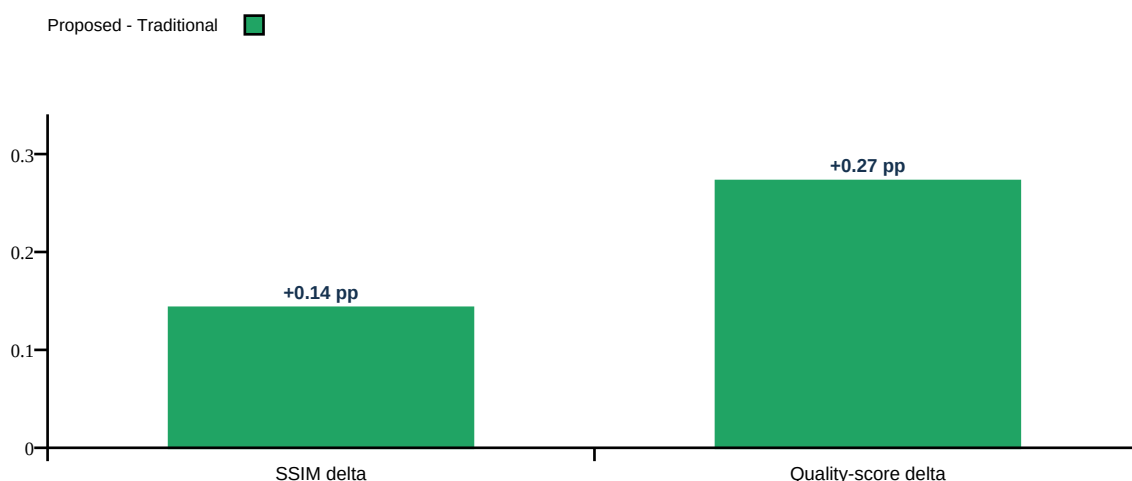
Metric	Original	Traditional	Proposed	Proposed delta
Contrast	0.1701	0.2153	0.2147	-0.0007
Ridge-Valley Clarity	8,226	13,274	13,016	-258.4
Sharpness (Laplacian)	1,519	2,429	2,383	-46.7
Edge Clarity (Sobel)	57.98	70.65	70.22	-0.43
SSIM (Structural Similarity)	1.000	0.979	0.980	+0.001

SSIM uses the original input as the 1.000 reference. It measures structural preservation and must not be interpreted as a percentage enhancement gain.

Enhancement Change vs Original (%)



Proposed Structural-Quality Advantage (percentage points)



Delta-chart interpretation: positive values favour Proposed Improved RHLT. For this image, the SSIM difference was +0.001 (+0.14 percentage points) and the balanced-quality difference was +0.0027 (+0.27 percentage points). These are small absolute advantages and do not establish statistical significance. Traditional RHLT may still produce marginally stronger enhancement measurements, while the Proposed method is selected when it preserves structure and passes the safety checks.

Visual Comparison



Original fingerprint

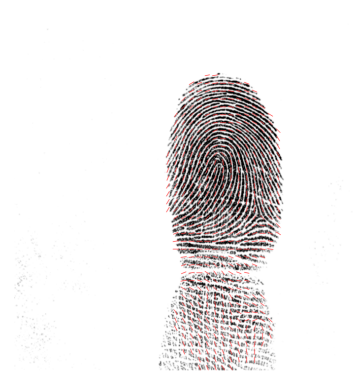


Traditional RHLT Baseline



Proposed Improved RHLT - Final
Selected

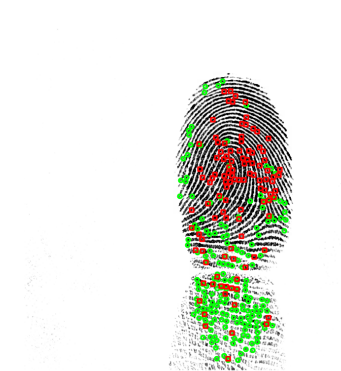
Diagnostic Outputs



Ridge orientation field



Ridge binary skeleton



Minutiae overlay

Detailed Metrics: Original vs Enhanced

Metric	Original	Enhanced	Change	Status
Contrast (RMS)	0.1701	0.2147	+0.0446	+
Sharpness (Laplacian)	1,519	2,383	+863.5	+
Edge Clarity (Sobel)	57.98	70.22	+12.24	+
Ridge-Valley Clarity	8,226	13,016	+4790.2	+
SSIM (Structure Similarity)	N/A	0.980	N/A	
PSNR (dB)	N/A	24.47	N/A	
MSE	N/A	232.17	N/A	
Foreground Coverage (%)	N/A	18.0	N/A	
Valid Orientation Blocks	N/A	450	N/A	
Mean Orientation Coherence	N/A	0.573	N/A	
Total Minutiae	N/A	298	N/A	
Ridge Endings	N/A	190	N/A	
Bifurcations	N/A	108	N/A	
Processing Time (ms)	N/A	2,250	N/A	